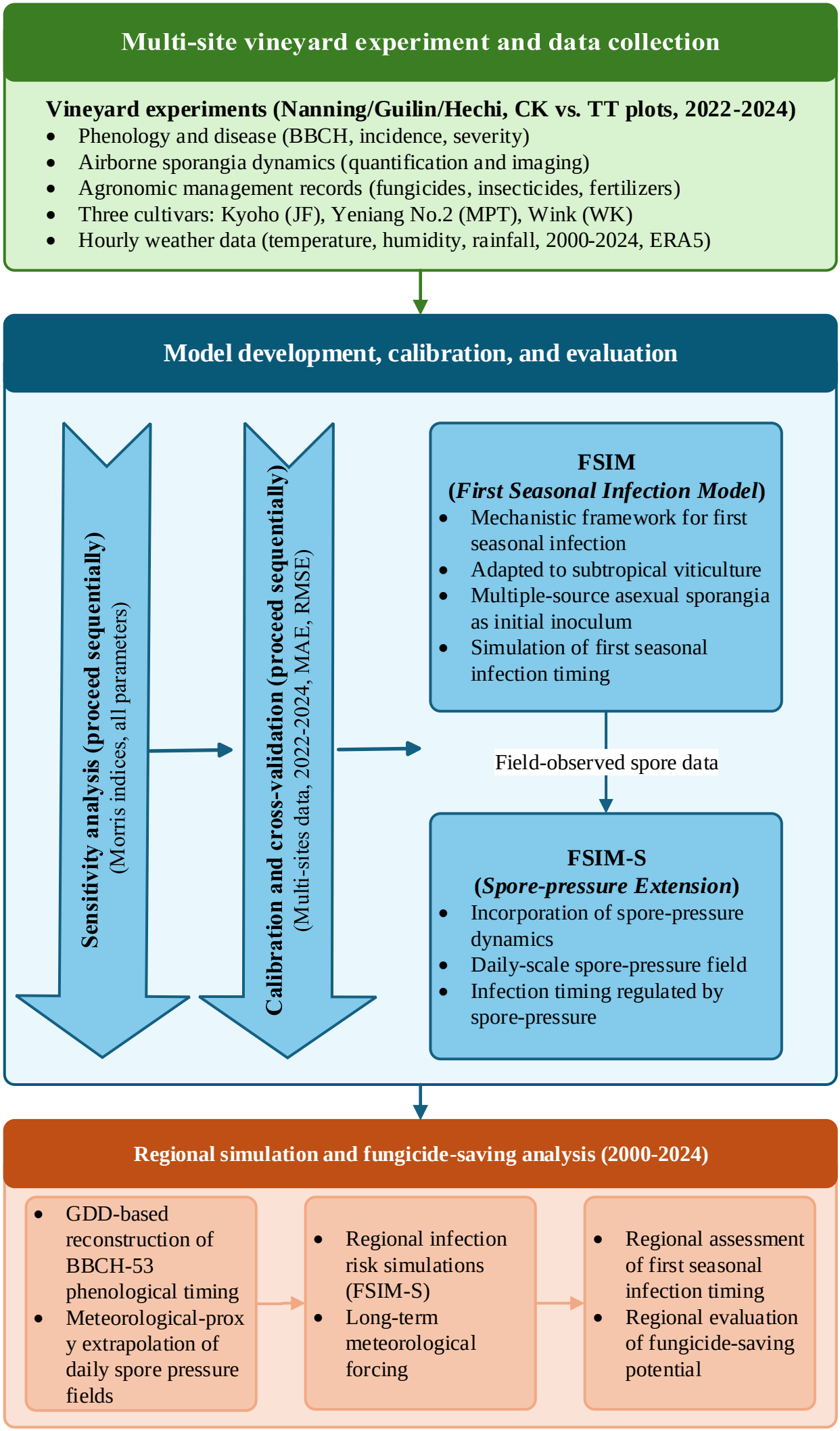

Multi-site vineyard experiment and data collection
Vineyard experiments (Nanning/Guilin/Hechi, CK vs. TT plots, 2022-2024)
Phenology and disease (BBCH, incidence, severity)
Airborne sporangia dynamics (quantification and imaging)
Agronomic management records (fungicides, insecticides, fertilizers)
Three cultivars: Kyoho (JF), Yeniang No.2 (MPT), Wink (WK)
Hourly weather data (temperature, humidity, rainfall, 2000-2024, ERA5)
Model development, calibration, and evaluation
Sensitivity analysis (proceed sequentially)
(Morris indices, all parameters)
Calibration and cross-validation (proceed sequentially)
(Multi-sites data, 2022-2024, MAE, RMSE)
FSIM
(First Seasonal Infection Model)
Mechanistic framework for first seasonal infection
Adapted to subtropical viticulture
Multiple-source asexual sporangia as initial inoculum
Simulation of first seasonal infection timing
Field-observed spore data
FSIM-S
(Spore-pressure Extension)
Incorporation of spore-pressure dynamics
Daily-scale spore-pressure field
Infection timing regulated by spore-pressure
Regional simulation and fungicide-saving analysis (2000-2024)
GDD-based reconstruction of BBCH-53 phenological timing
Meteorological-proxy extrapolation of daily spore pressure fields
Regional infection risk simulations (FSIM-S)
Long-term meteorological forcing
Regional assessment of first seasonal infection timing
Regional evaluation of fungicide-saving potential